

The effects of Medium Chain Triglycerides on sports performance and energy expenditure in healthy young adults - A scoping review

Student ID: 220070511

Assignment: Systematic Review Part 2

Name of module: Literature review

Date of submission: 27/04/2026

Submitted in part fulfilment of the intercalated BSc in Sports and Exercise Medicine

Advisors: Dr Aivaras Ratkevicius

Word Count: 3994 (excluding page 1, titles, references, figures and tables)

Title:

The Effects of Medium Chain Triglycerides Supplementation on Exercise Performance and Metabolism in Young Healthy Adults: A scoping review

Abstract:

Background:

Medium-chain triglycerides (MCTs) have been put forward as a potential nutritional supplement to increase sports performance and metabolic output through increasing fat oxidation. However, the existing literature is limited, and the findings are often conflicting. This scoping review aims to lay out the available evidence on the effects of MCFA supplementation on exercise performance and metabolism in healthy young adults.

Method:

A scoping review was carried out following the PRISMA-ScR guidelines. The databases included were PubMed, ScienceDirect and Embase. The studies were considered if they included young healthy participants, MCT supplementation and examined outcomes regarding exercise performance and metabolism. Both acute and chronic supplementation regimes were considered. After screening, the data was extracted using a template created on Covidence.

Results:

From the 1093 studies, 16 met the inclusion criteria after screening. Most of these 16 studies were cross-over trials in trained or healthy adults. The findings illustrated MCT supplementation increases ketone production and fatty-acid oxidation, however the results regarding exercise performance were mixed. Doses that were high and given acutely showed decreased performance due to gastrointestinal discomfort. Whereas chronic supplementation showed some evidence of increased endurance and energy output. There wasn't any consistent evidence regarding increased energy expenditure and sparing of carbohydrates.

Discussion:

The conflicting results may be a result of the heterogenous nature of the studies included regarding the design, participant characteristics, amount and timing of the dosages. While there may be some changes in

substrate utilisation, this does not necessarily translate to improved performance.

Conclusion:

MCT increases fat oxidation during exercise, but physical performance is inconsistent. There needs to be further research regarding the best dosage and its long-term effects.

Keywords:

Medium-chain triglycerides; exercise performance; fat oxidation; metabolism.

Introduction:

Rationale:

Substrate utilisation is a cornerstone of endurance exercise performance. Medium-chain triglycerides (MCTs) have recently gained popularity due to their rapid absorption and oxidation compared to long-chain triglycerides (LCTs). MCTs are broken down into medium-chain fatty acids (MCFAs) that enter the portal vein directly which is faster compared to LCFAs that are absorbed into the lymphatic circulation initially[1]. MCFAs then enter mitochondrial pathways by passing through the inter membrane space without the reliance of the carnitine shuttle. This provides a readily available energy source during exercise[2][3].

These characteristics have led to the hypothesis that MCT supplementation could be an effective nutritional tool to better exercise performance through its metabolic qualities. Theoretically, they provide an advantage through increased fat oxidation, reducing reliance on carbohydrate metabolism, and sparing muscle glycogen[4]. However, the existing evidence remains inconsistent. Some studies report elevated thermogenesis, increased ketone production and better endurance performance, whereas others demonstrate no changes or even worse performance often due to gastrointestinal distress.

A key aspect within the existing evidence is the heterogeneity across the studies including differences in the dose of MCTs, acute versus chronic strategies, participant characteristics and the measured outcome measures. This variation makes it challenging to note direct comparisons and thus a definitive conclusion cannot be made regarding the potential of MCT as an exercise supplement.

With this inconsistency being the primary reason, a scoping review was regarded the most appropriate type of review[5]. This will allow the mapping of the existing findings, identify any patterns and disparities, and highlight any gaps for future research. The aim of this review is to examine the effects of MCT supplementation on exercise performance and metabolic changes in young healthy adults.

Objectives:

Lay out the existing evidence on the effects of medium-chain triglycerides (MCTs) on exercise performance in young healthy adults

Examine the metabolic effects of MCT supplementation including energy expenditure and substrate utilisation.

Highlight limitations, inconsistencies and gaps within the existing literature for future research.

Methods:

Protocol and registration:

The protocol for this scoping review has not been registered. However, the review was conducted conforming to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses Extension for Scoping Reviews (PRISMA-SCR) guidelines[6].

Eligibility criteria:

The eligibility criteria were created using the Population-Concept-Content framework to provide a general structure for the criteria[7]. Studies were either included or excluded based on the characteristics of the participants, type of intervention and the outcomes that were measured. Table 1 outlines the criteria.

	Inclusion criteria	Exclusion Criteria
Population	Young healthy adults (18-35 years old); physically active or trained individuals; male and female participants	Children (<18 years old), older adults (>65 years old); clinical populations (e.g., metabolic, cardiovascular, endocrine conditions); animal or in vitro studies
Concept	Medium-chain fatty acid (MCFA) or medium-chain triglyceride (MCT) supplementation; acute or chronic interventions; oral-ingestion	Long-chain fatty acid supplementation; mixed interventions where MCT/MCFA effects cannot be isolated; non-oral (e.g., intravenous) administration
Context	Exercise or metabolism-related settings; testing during rest or exercise	Studies not related to exercise or metabolic settings
Outcomes	Exercise performance (e.g., time-to-exhaustion, endurance capacity, time trials); metabolic outcomes (e.g., substrate utilisation, fat oxidation, energy expenditure, VO ₂)	Outcomes unrelated to exercise or metabolic performance
Study characteristics	Experimental (e.g., randomised controlled trials, crossover trials) and observational studies; English	Reviews, non-english publication

Table 1 – inclusion and exclusion criteria for the study

The search for relevant studies was carried out on three different electronic databases including PubMed, ScienceDirect, and Embase. The final search was conducted on the 11th of April 2026. The reference list of the included studies was also screened for additional studies that could be relevant.

A refined search strategy was used to identify relevant studies to begin the screening process. It was developed using keywords and variations of those keywords combined with Boolean operators AND and OR. The search strategy was tailored differently for each database due to differences in the database's advanced search and filter options. Table 2 outlines the strategies for all the databases searched:

Database	Search Strategy
PubMed	("medium chain fatty acid*" OR "medium chain triglyceride*" OR "medium chain fatty acid*" OR MCT OR MCFA OR "octanoic acid" OR "decanoic acid" OR "caprylic acid" OR "capric acid" OR medium chain triglyceride[Title] OR medium-chain triglyceride[Title] OR medium chain fatty acid[Title] OR medium-chain fatty acid[Title] OR octanoic acid[Title] OR decanoic acid[Title] OR caprylic acid[Title] OR capric acid[Title] OR MCFAs[Title] OR MCFA[Title] OR MCT[Title]) AND ("energy metabolism"[Mesh] OR "oxygen consumption"[Mesh] OR "calorimetry, indirect"[Mesh] OR "physical endurance"[Mesh] OR "athletic performance"[Mesh] OR "energy expenditure" OR "resting metabolic rate" OR "metabolic rate" OR "substrate

	oxidation" OR "fat oxidation" OR VO2 OR "indirect calorimetry" OR "exercise performance" OR "physical performance" OR endurance OR "aerobic capacity" OR "sports performance" OR "time trial" OR "exercise test" OR exercise[MeSH] OR exercise OR training OR sport* OR "physical activity") Clinical Trial, Observational Study, Randomized Controlled Trial, English, Humans, Adult: 19+ years.
ScienceDirect	Article: ("energy expenditure" OR "energy metabolism" OR "fat oxidation" OR "exercise performance" OR endurance OR "sports performance" OR "aerobic capacity" OR "VO2 Max" OR "Energy expenditure") Title: ("medium chain triglyceride" OR "medium chain fatty acid" OR MCT OR MCFA) Title key terms: ("medium chain triglyceride" OR "medium chain fatty acid" OR MCT OR MCFA)
Embase	('medium chain triglyceride' OR 'medium chain fatty acid' OR MCT OR MCFA) AND ('energy expenditure' OR 'energy metabolism' OR 'fat oxidation' OR 'oxygen consumption' OR 'exercise performance' OR 'sports performance' OR endurance OR 'aerobic capacity') AND (supplement OR ingestion OR intake OR diet OR experiment) AND ('young adult' OR 'college student' OR 'healthy adult' OR athlete OR athletes OR "physically active") AND [humans]/lim AND [english]/lim

Table 2 – Search strategies used for study selection

Study selection:

The screening process was carried out on Covidence by two researchers and consisted of three stages. This was done independently and if there were any conflicts within the studies, the researchers would have a discussion to reach a final verdict. The first stage of the screening is removing duplicates within the imported studies. This is done automatically by the Covidence software[8].

Following this, the remaining studies are screened for their title and abstract to assess relevance and alignment with the eligibility criteria. The third stage of screening includes sourcing the full-text for the remaining studies and examining closely if the study is eligible under the criteria previously stated[9].

Data items and charting process

Data was charted using an extraction tool in the form of a table created on Covidence. The template was designed specifically for this review and aims to systematically capture the relevant information from each of the included studies that is related to the objectives.

The extracted data included identification details of the study (e.g., citation, country and design type), participant characteristics (e.g., age, sex, training status and sample size) and data regarding the concept and context of the study. This consisted of details of the medium-chain triglyceride (MCT) supplementation protocols including the dosage, method of administration, timing as well as the exercise protocol[10].

The outcomes of the studies were also extracted including both performance-related outcomes and metabolic responses, alongside the measurement tools used within the studies. This included variables such as time-to-exhaustion, time-trial performance, VO2 max, energy expenditure, fat oxidation and substrate utilisation. In addition, key findings, authors' conclusions, and reviewer interpretations relating to study quality, limitations and its relevance to the scoping review[10]. After individual screening of the studies, the two researchers cross-checked for accuracy and completeness of the extraction. Figure 1 shows the extraction template.

Study Identification	Intervention (I)	Outcomes
Citation details 	Type of MCFA used 	Primary outcomes
Country of study Title of paper / abstract / report that data are extracted from 	Supplement dose & duration 	Secondary outcomes
Study design 	Timing / administration 	Measurement tools
Population (P) Participant characteristics 	Comparator 	Results / Key Findings Summary of main findings
Health/Fitness status 	Context (C) Setting 	Authors' conclusions
Sample size 	Protocol 	Quality / Limitations Reviewer notes
		Gaps / Relevance to Review Reviewer summary

Figure 1

Critical appraisal:

For the studies that were considered eligible for the review, a formal critical appraisal was not done. This was primarily due to the aim of the review which is to map out the existing literature rather than assess the quality of the methods used in these individual studies. This is consistent with the PRISMA guidance on scoping review[6].

Synthesis of results:

The results from the extracted data were synthesised through a descriptive approach in line with the objectives of the scoping review. The findings were categorised into four main thematic groups. Table 3 summarises the four themes explored.

Theme	Variables examined
Exercise Performance	VO2 max, time trial, time to exhaustion
Substrate utilisation	Glucose oxidation, fat oxidation, RER
Metabolic responses	Energy expenditure, thermogenesis, ketone concentration
Gastrointestinal discomfort	Reported GI distress

Table 3 – Themes that were explored in the included studies

Data were summarised to identify any potential patterns, differences and similarities. The variations in supplementation doses, participants and duration were also considered. Quantitative analysis was not carried out due to the vast heterogeneity of the studies including its designs, outcome measures and protocols.

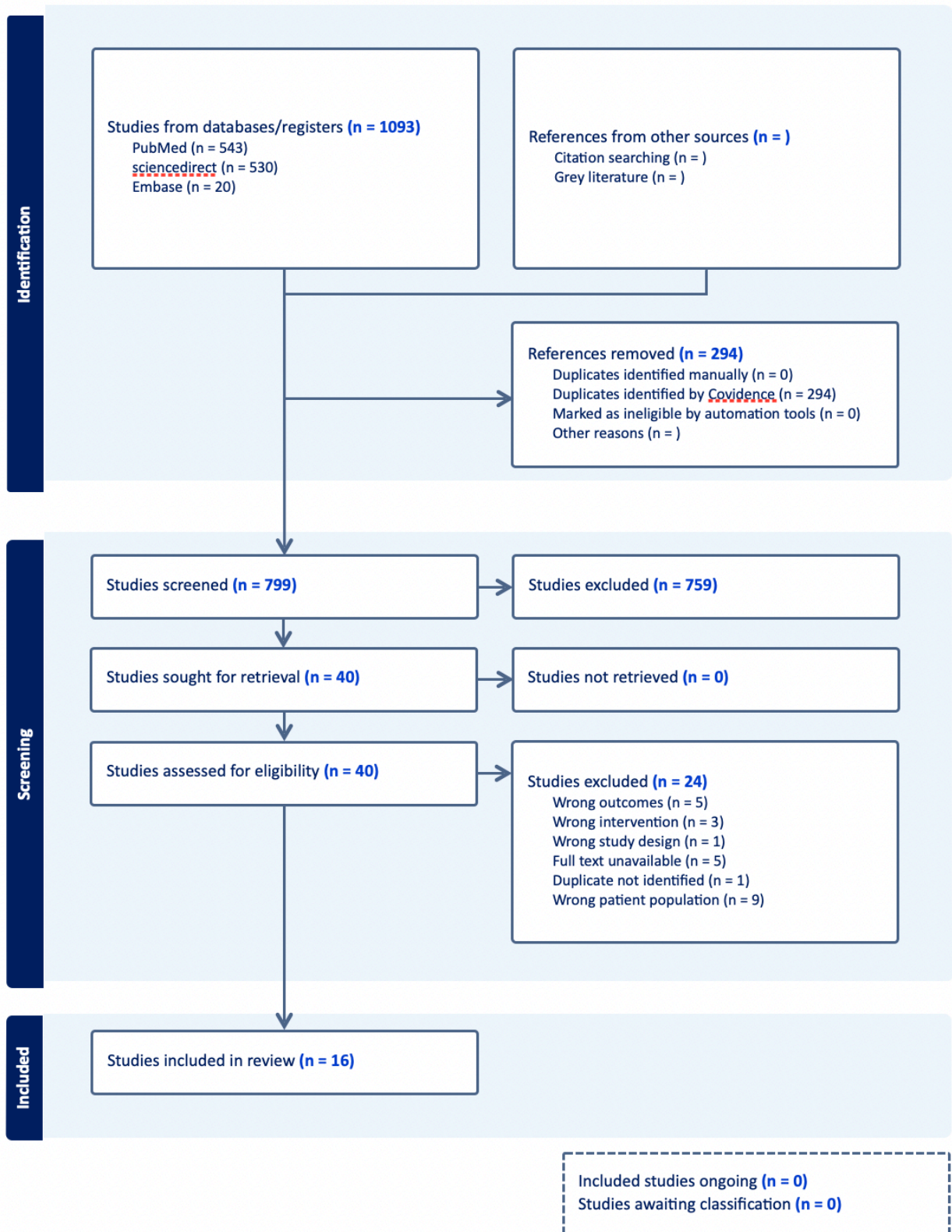
Results:

Selection of sources of evidence

A total of 1093 studies were identified through the search strategy. They were then imported into Covidence where 294 were removed due to being duplicates. Following this stage, the title and abstract of the remaining 799 studies were screened. Of these, 40 studies were assessed to be eligible.

The full texts for these 40 articles were then imported to be studied for further screening. A total of 16 studies met the inclusion criteria and were included to be in the final review. The other 24 studies were excluded. Of the 24, 9 of them were excluded due to the participant characteristics not aligning with the population this review is interested in[11-19]. 5 studies were excluded due to a lack of access to the full text[20-24]. 5 studies were excluded due to the measured outcomes not being relevant to the review's aims[25-29]. 3 studies were excluded due to the intervention not being relevant[30-32]. 1 study was excluded because of its study design[33]. 1 study was excluded as it was a duplicate of another study that was already included and was not identified in the initial screening[34]. Figure 2 illustrates the study selection process in a PRISMA flow diagram.

Dietary supplements of medium chain fatty acid and metabolism at rest and during exercise.



21st April 2026



Figure 2 - Prisma flow diagram

Characteristics of included studies:

Study designs:

16 studies were included in this scoping review. All of them were randomised crossover trials. This design allows for the participants to be their own control and assess comparisons[34]. This is particularly important in studies relating to exercising and metabolic factors since there is high variability between individual physiologies[35]. This large proportion of crossover trials illustrates the regulated, laboratory-based aspect of this field.

Participants:

All participants from the included studies were healthy young adults between the ages 18-40. 8 of the studies focused on endurance athletes such as cyclists and runners to look at metabolic changes at a very high level of physical activity while the other 8 focussed on recreationally active individuals. The sample sizes across the 16 studies were quite small ranging from 6 to 16 participants. The participant pool consisted of only males in most of the studies. There were some studies which had a mix of both males and females. This lack of female representation is potentially a gap within the literature.

Intervention:

The protocols across the studies used Medium-chain Triglycerides. However, there was variation within the supplementation procedures in regard to the dose, timing and duration of the regime. 6 studies aimed to examine the effects of chronic MCT oil supplementation over several days. 2 studies examined the effects after several days of adapting to lower doses combined with a higher acute intake before testing. The other 8 studies focused solely on the acute effects of MCT supplementation on metabolic markers and exercise.

The timing of the supplementation was also varied within the studies. For some, it was taken right before exercise, while in other studies, it was incorporated into their daily diets. This vast heterogeneous nature of supplementation regimes of these studies has led to the contradicting results in the reported outcomes resulting in the literature being inconsistent.

Context:

Of the studies included, 10 looked at the effects of MCT on exercise thus the protocols consisted of endurance-type exercise testing. Cycling was the most common modality due to the equipment being able accurately measure performance variables and standardise workload. All the exercise protocols investigated the endurance of the participants by having prolonged sub maximal exertion during the exercise. The other 6 studies did not employ exercise protocols but rather resting metabolic measurements.

Furthermore, 15 out of the 16 studies were done in a controlled laboratory setting allowing for consistency and higher accuracy within the results while only 1 study was done in free-living conditions[37].

Outcomes:

All of the included studies reported metabolic variables such as energy expenditure, fat oxidation and substrate utilisation. These variables were examined in two main ways: using gas analysis and blood markers. These measures give a reliable insight on how MCT oil changes the physiology of energy production within the body.

Exercise performance outcomes were reported in the 10 studies that utilised exercise protocols. They commonly reported time-trials, time-to-exhaustion and endurance capacity. This allows for the effect of MCT oil on endurance. Another outcome that was measured were any symptoms of GI distress which could possibly lead to decreased exercise performance.

Results of individual sources of evidence:

The data was initially obtained using the extraction template outlined earlier. To simplify synthesis, less relevant characteristics that do not answer the objectives of the review were removed. Table 4 below illustrates the relevant information from each study.

In summary, the included studies reported a range of findings relating to exercise performance and metabolic responses following MCT supplementation. While some studies demonstrated increases in fat oxidation and energy expenditure, the majority reported no significant improvements in exercise performance. A number of studies also identified negative outcomes, including gastrointestinal discomfort and reduced performance under certain conditions.

Citation	Study design	Participants	Intervention	Context	Outcomes	Results
Alexandrou et al. (2007)	Randomized crossover feeding study (7-day)	Healthy, recreationally active women; n = 8	MCT (~25% energy, 7 days) vs isocaloric LCT diet	Laboratory-based; resting and postprandial (no exercise)	RMR, energy expenditure, fat & CHO oxidation, RQ (indirect calorimetry)	Increased Fat oxidation, Decreased CHO oxidation, Decreased RQ with MCT; no change in RMR or total energy expenditure
Angus et al. (2000)	Randomized, double-blind crossover trial	Endurance-trained male cyclists/triathletes; n = 8	CHO vs CHO + MCT (4.3% MCT + 6% CHO) vs placebo, consumed during exercise	Laboratory 100-km cycling time trial (~75% VO ₂ peak)	Time-trial performance, substrate oxidation, blood markers	CHO improved performance vs placebo; CHO + MCT showed no additional benefit; no change in fat oxidation vs CHO; GI distress reported with MCT
Hill et al. (1989)	Randomized, double-blind crossover overfeeding	Healthy, non-athletic males; n = 10	MCT (~40% dietary fat, overfeeding) vs isocaloric LCT diet	Inpatient metabolic ward; resting/postprandial (no exercise)	Energy expenditure, TEF, substrate oxidation, ketones	Increased thermic effect of food and energy expenditure

	study (7-day)					with MCT; increased fat oxidation and ketones; no change in resting metabolic rate; GI discomfort reported
Jeukendrup et al. (1996)	Randomized, double-blind crossover trial	Endurance-trained male cyclists/triathletes; n = 9	CHO vs CHO + MCT (~29 g during exercise)	Laboratory cycling (180 min, ~57% VO ₂ max)	Substrate oxidation, glycogen use, blood markers	No effect of MCT on CHO oxidation, glycogen breakdown, or fat oxidation; ketones increased with MCT; no improvement in substrate utilisation
Jeukendrup et al. (1995)	Randomized, double-blind crossover trial	Endurance-trained male cyclists/triathletes; n = 8	MCT (~29 g) ± CHO during exercise	Laboratory cycling (180 min, ~57% VO ₂ max)	MCT oxidation, substrate utilisation, blood markers	MCT highly oxidised (~70%) but contributes minimally to total energy (3–7%); no effect on VO ₂ , RER, or substrate utilisation; ketones increased with MCT

Jeukendrup et al. (1998)	Randomized crossover trial	Highly trained male cyclists; n = 7	CHO vs CHO + MCT (~85 g) vs MCT vs placebo	Laboratory cycling (2 h at ~60% $\text{VO}_{2\text{max}}$ + time trial)	Performance, substrate oxidation, blood markers, GI symptoms	No improvement in performance with MCT; MCT alone decreased performance; no effect on CHO or fat oxidation vs CHO; ketones increased; significant GI distress with MCT
Kanta et al. (2025)	Randomized crossover trial	Healthy, recreationally active men; n = 16	Acute (35 g) and chronic (10–30 g/day, 8 days) MCT vs LCT	Laboratory; resting conditions (no exercise)	EPO, ketones, hematological markers	increased Ketone bodies with MCT; no acute change in EPO; chronic MCT increased basal EPO; no change in hemoglobin or hematocrit
Kasai et al. (2002)	Double-blind, randomized crossover study	Healthy adults (male and female); n = 16	Acute MCT (5–10 g) vs LCT	Laboratory; resting postprandial conditions (no exercise)	Diet-induced thermogenesis, oxygen consumption, respiratory quotient	MCT increased diet-induced thermogenesis and oxygen consumption compared to LCT; respiratory quotient decreased, indicating greater fat oxidation
Nosaka et al. (2009)	Randomized, double-blind crossover trial	Recreationally active young adults; n = 8	MCT (6 g/day for 14 days) vs LCT	Laboratory cycling (moderate intensity followed by high-intensity to exhaustion)	Time to exhaustion, lactate, RPE, substrate oxidation	Time to exhaustion increased with MCT; lactate and perceived exertion decreased; no significant change in VO_2 , RER, or

						substrate oxidation
Nosaka et al. (2018)	Randomized, double-blind crossover trial	Female recreational athletes; n = 8	MCT (6 g/day, 2 weeks) + carbohydrate vs carbohydrate alone	Laboratory cycling (moderate intensity followed by exercise to exhaustion)	Time to exhaustion, substrate oxidation, RER, blood markers, RPE	Fat oxidation increased and carbohydrate oxidation decreased with MCT; time to exhaustion increased; RER decreased indicating greater fat utilisation; perceived exertion decreased; no change in blood glucose or lactate
Oöpik et al. (2001)	Randomized, double-blind crossover trial	Highly trained male endurance runners; n = 7	MCT (~34 g/day, 7 days) vs placebo	Laboratory treadmill running (~80% VO ₂ max to exhaustion)	Time to exhaustion, ketones, blood markers	No improvement in performance; performance decreased with MCT; ketone levels increased; no meaningful change in glucose or lactate; gastrointestinal discomfort reported

Papamandjaris et al. (1999)	Randomized crossover feeding study (14 days)	Randomized crossover feeding study (14 days)	MCT-enriched diet (~26% MCFA) vs LCT diet	Free-living with controlled diet (no exercise)	Total energy expenditure, BMR, TEF, activity-induced expenditure	No difference in total energy expenditure between MCT and LCT; temporary increase in BMR not sustained; no changes in TEF or activity-related expenditure
Thorburn et al. (2006)	Randomized, double-blind triple crossover trial	Highly trained male cyclists; n = 9	Chronic MCFA diet (~2 weeks) + acute MCFA ingestion vs CHO control	Laboratory endurance cycling (3 h + sprint test)	Performance, substrate oxidation, GI distress, blood markers	Performance decreased with MCT; adaptation reduced gastrointestinal distress but did not improve performance; high MCT oxidation observed; slight decrease in carbohydrate oxidation; GI distress negatively impacted performance
Van Wymelbeke et al. (2001)	Randomized crossover laboratory study	Healthy young men; n = 10 (9 analysed)	Acute MCT (~35 g) vs LCT vs carbohydrate meal	Acute MCT (~35 g) vs LCT vs carbohydrate meal	Substrate oxidation, energy expenditure, food intake, blood markers	MCT increased fat oxidation and decreased carbohydrate oxidation; no increase in energy expenditure; reduced subsequent food intake; ketones increased; no effect on satiety duration

Van Zyl et al. (1996)	Randomized crossover trial	Highly trained male cyclists; n = 6	MCT (~86 g) vs CHO vs CHO + MCT during exercise	Laboratory cycling (2 h at ~60% VO ₂ max + 40-km time trial)	Time trial performance, substrate oxidation, blood markers	MCT alone decreased performance; CHO + MCT showed a small improvement compared to CHO; fat oxidation and ketones increased; carbohydrate oxidation decreased
Vistisen et al. (2003)	Randomized crossover trial	Highly trained male cyclists; n = 7	Structured MCT (MLM) vs CHO during exercise	Laboratory cycling (3 h at ~55% VO ₂ max + time trial)	Time trial performance, substrate utilisation, plasma fatty acids	No difference in performance between conditions; no detectable increase in circulating medium-chain fatty acids; slight increase in fat oxidation; no meaningful change in overall metabolism; gastrointestinal symptoms reported

Table 4

Synthesis of Results:

Exercise Performance:

A large proportion of the included studies in this review reported no significant increase in exercise performance after medium-chain triglyceride (MCT) supplementation. This was seen even with the differences in dosage, duration, or exercise protocol. Both acute and long-term MCT intake did not show enhanced performance outcomes such as time trial completion or power output. The findings are consistent across studies involving endurance- trained athletes[39,43,48,50,53]. In some of the studies MCT supplementation was associated with a reduction in performance compared to carbohydrate- based conditions. This was particularly seen when administered in higher doses[43,48,50].

Whereas a few studies reported slight increments in exercise capacity following MCT supplementation. These improvements were mainly seen in recreationally active young adults rather than trained athletes and were typically measured using time-to-exhaustion protocols rather than performance-based outcomes such as time trials[46,47]. Additionally, one of the studies reported a small improvement in performance when MCT was ingested with carbohydrate. However, this effect was very small and not consistently replicated in later research[52].

Throughout all the included studies, carbohydrate intake by itself consistently demonstrated clear ergogenic benefits, enhancing performance compared to both placebo and MCT conditions[39]. When MCT was combined with carbohydrates, it did not provide any further enhancement in performance in most cases, suggesting that MCT does not offer additional benefits beyond established nutritional strategies. Furthermore, many participants experienced gastrointestinal discomfort in studies involving MCT supplementation, particularly at higher doses. This may have negatively affected performance outcomes[43,48,50].

Overall, the evidence illustrates that MCT supplementation does not enhance exercise performance and could in fact negatively affect it. While limited evidence suggests potential benefits under specific conditions, these findings are inconsistent and only seen in recreationally active participants, meaning that it cannot be applied to the general public.

Substrate utilisation:

In the 16 included studies, MCT supplementation was consistently associated with greater fat oxidation and reduced reliance on carbohydrates for energy displaying a shift in substrate utilisation. Many studies conducted under both resting and exercise conditions reported decreases in respiratory quotient following MCT ingestion, indicating increased fat utilisation[45,47,51]. This was the case in both acute and short-term protocols, suggesting a consistent metabolic response to MCT intake.

However, despite this shift in substrate utilisation, most of the exercise-based studies reported no meaningful changes in overall carbohydrate oxidation or muscle glycogen utilisation when MCT was co-ingested with carbohydrate[41,43]. Likewise, no significant differences in respiratory exchange ratio or oxygen consumption were seen between MCT and carbohydrate conditions in several trials, indicating that the overall contribution of MCT to energy metabolism during exercise remains limited[41,53].

Further investigations demonstrated that although MCTs are oxidised quickly, they only contribute a small proportion of the overall total energy expenditure during exercise[42]. This little contribution suggests that, while MCTs can be utilised as a fuel source, they do not alter the overall balance of substrate utilisation during exercise to a large degree.

Overall, while MCT supplementation promotes a shift towards increased fat oxidation, this effect is small and does not cause meaningful changes in total substrate utilisation during exercise.

Metabolic Response:

Throughout the studies, there was a clear association between MCT supplementation and increase in circulating ketone bodies across both resting and exercise conditions, reflecting enhanced hepatic oxidation of medium- chain fatty acids. Many studies reported significant increments in ketone concentrations in the

blood after MCT intake. This was the case in both acute and short-term protocols[42,44,48,53]. Even with this increase in ketone availability, it did not improve exercise performance.

In contrast, the total energy expenditure and thermogenesis did show meaningful changes. Some studies displayed increases in postprandial energy expenditure and thermic effect of food following MCT consumption[40,45], suggesting a potential role of MCT in enhancing metabolic rate. However, other studies reported no significant differences in total energy expenditure. This was particularly the case over longer intervention periods, suggesting that these effects may not be sustained[38,49].

Additionally, wider physiological responses to MCT supplementation appeared limited. While ketone production was consistently increased throughout the studies, there were generally no significant changes in key metabolic markers such as lactate or blood glucose during exercise[43,48]. Moreover, newer evidence suggests that MCT ingestion may influence other physiological pathways, such as erythropoietin production; however, the relevance of these findings to exercise performance remains unclear[44].

Overall, MCT supplementation induces measurable metabolic responses, particularly increased ketone production and, in some cases, elevated energy expenditure. However, these effects are inconsistent and do not appear to result in meaningful changes in exercise-related metabolic function.

Gastrointestinal Discomfort:

Gastrointestinal discomfort was often reported across studies investigating MCT supplementation. This was particularly the case in those involving higher doses or ingestion during exercise. Symptoms such as nausea, bloating, and general gastrointestinal distress were observed in many performance-based studies and were often correlated with reduced exercise capacity[43,48,50]. In some cases, this adverse effect indicated impaired performance, especially when large quantities of MCT were consumed acutely prior to prolonged exercise.

Attempts to reduce gastrointestinal symptoms through dietary adaptation showed limited success. One study reported that a period of MCT adaptation reduced the severity of gastrointestinal discomfort. However, this did not result in improvements in exercise performance[50]. This suggests that although tolerance to MCT may improve over time, underlying issues related to digestion and absorption may still influence exercise outcomes.

Furthermore, gastrointestinal discomfort appeared to be dose-dependent, with higher intakes of MCT more likely to cause more severe symptoms. Studies using lower doses reported less adverse effects, although these were also less likely to demonstrate any measurable performance or metabolic benefit[46,47]. This highlights a potential trade-off between tolerability and efficacy, where doses sufficient to elicit metabolic changes may also increase the likelihood of gastrointestinal distress.

Overall, gastrointestinal discomfort represents a consistent and important factor associated with MCT supplementation, particularly in exercise settings. Its presence may contribute to the lack of observed performance benefits and, in some cases, may directly impair exercise capacity.

Discussion

The aim of this scoping review was to examine the effects of medium-chain triglyceride (MCT) supplementation on exercise performance and associated metabolic responses in healthy young adults. Overall, even with significant measurable changes in metabolism, the reported outcomes indicate that MCT

supplementation does not increase exercise performance. Throughout the studies, increases in ketone production and fat oxidation were consistently seen following MCT intake; however, these changes did not translate into improvements in endurance performance or exercise capacity. On the other hand, carbohydrate supplementation alone was shown to provide clear ergogenic benefits, and the addition of MCT did offer any further improvements. Moreover, some studies reported negative outcomes in relation to MCT ingestion, including gastrointestinal discomfort and, in some cases, decreased performance. These findings suggest that while MCT supplementation influences metabolic pathways, but it does not provide a meaningful ergogenic benefit in the context of exercise performance.

Mechanism:

An important reason for the lack of enhanced performance observed with MCT supplementation lies in the limited contribution of energy derived from MCT to the overall exercise metabolism. Despite being rapidly absorbed and oxidised, evidence suggests that MCT contribution to total energy expenditure during exercise is relatively small[42]. This shows that, even with their rapid availability, MCTs do not supply sufficient energy to meaningfully influence performance outcomes. Moreover, studies have shown that the availability of medium-chain fatty acids in the blood may be limited during exercise, which could restrict their uptake and utilisation by muscles[53]. As a result, the hypothetical advantage of MCT as a readily available source of energy does not appear to translate into a benefit during actual endurance exercise.

Furthermore, while MCT supplementation causes a shift towards increased fat oxidation and ketone production, these changes do not appear to significantly alter overall substrate utilisation during exercise. Carbohydrate oxidation remains the primary energy pathway, especially at moderate to high exercise intensities, where energy demands are high and rapid ATP production is needed. Consequently, any increase in fat oxidation associated with MCT intake is insufficient to reduce reliance on carbohydrate metabolism or meaningfully spare muscle glycogen stores. This is supported by findings showing no significant differences in carbohydrate oxidation or glycogen utilisation between MCT and carbohydrate conditions[41,43]. When they are combined, the findings suggest that although MCT supplementation induces metabolic changes, these differences are not big enough to influence exercise performance.

Inconsistencies:

Even with the overall lack of exercise performance benefit, there were some inconsistencies in findings that was seen across the included studies. This may be due to the differences in participant characteristics, supplementation and exercise protocols. Studies that reported small improvements in exercise capacity were primarily conducted in recreationally active young adults rather than trained athletes[46,47]. This suggests that training status may influence the effects of MCT supplementation, with less-trained individuals benefitting more due to lower metabolic efficiency. On the other hand, highly trained athletes, who have adapted to have well-developed oxidative capacity and efficient substrate utilisation, may be less likely to benefit from alternative fuel sources such as MCT.

The vast differences in dosage and supplementation strategies may also contribute to inconsistent findings. Studies using lower doses of MCT tended to report little to no metabolic or performance effects, whereas higher doses were more likely to induce gastrointestinal discomfort, potentially impairing performance[43,50]. This suggests a tight margin between ineffective and intolerable dosing, which limits the practical application of MCT supplementation. Furthermore, the different exercise protocols may

influence reported outcomes. In particular, the use of time-to-exhaustion tests versus time-trial performance measures. Time-to-exhaustion tests are known to be more variable and may overestimate small physiological changes, which could explain why some studies reported improvements under these conditions[46].

These factors highlight the heterogeneity within the literature and suggest that any potential benefits of MCT supplementation are likely to be context-specific and cannot be applied to the general populations or exercise conditions.

Strengths Limitations and further research:

This scoping review gives an overview of the existing evidence on MCT supplementation and exercise performance, highlighting both consistencies and areas of uncertainty. However, there are some limitations within the studies included. Most of the studies had small sample sizes, which may reduce statistical power and limit the generalisability of findings. Furthermore, most of the studies were carried out in controlled laboratory settings. Although this allows for variables to be controlled easily, it may not accurately reflect real-world environments. There was also a lack of diversity within study populations. Most studies exclusively had male participants or small, homogeneous groups. This limits the ability to apply the findings to broader athletic populations.

Furthermore, considerable heterogeneity was observed across studies in terms of supplementation protocols, including differences in MCT dosage, duration of intake, and timing relative to exercise. This variation makes direct comparison between studies challenging and may contribute to the inconsistent findings reported within the literature. Future research should aim to address these limitations by employing larger, more diverse participant samples and standardised supplementation protocols. In particular, further investigation into the long-term effects of MCT supplementation and its impact under real-world training conditions may provide great insight into its potential role in sports nutrition.

Conclusion

This scoping review assessed the effects of medium-chain triglyceride (MCT) supplementation on exercise performance and metabolic responses in healthy young adults. The findings show that, although MCT supplementation consistently alters metabolic processes by increasing fat oxidation and ketone production, these changes do not translate into improvements in exercise performance. Across the literature, carbohydrate remains the primary and most effective fuel source for endurance exercise, with MCT providing no additional ergogenic benefit when taken together. In some cases, MCT supplementation was associated with negative outcomes, including gastrointestinal discomfort and reduced performance, particularly at higher doses.

Overall, the current evidence does not support the use of MCT as an effective ergogenic aid for improving exercise performance. While some context-specific benefits have been reported, these findings are limited and not generalisable. Future research should focus on standardised protocols and diverse populations to further clarify the role of MCT in sports nutrition.

References:

Jia M, et al. Dietary fatty acids activate or deactivate brown and beige fat. *Biochim Biophys Acta Mol Cell Biol Lipids*. 2023. Available from: <https://www.sciencedirect.com/science/article/abs/pii/S0024320523006136>

Roopashree PG, Shetty SS, Kumari NS. Effect of medium chain fatty acid in human health and disease. *J Funct Foods*. 2021;87:104724. doi:10.1016/j.jff.2021.10472

Pereyra AS, McLaughlin KL, Buddo KA, Ellis JM. Medium-chain fatty acid oxidation is independent of L-carnitine in liver and kidney but not in heart and skeletal muscle. *Am J Physiol Gastrointest Liver Physiol*. 2023;325(4):G287–G294. doi:10.1152/ajpgi.00105.2023

Impact of MCT oil and caffeine on substrate metabolism during submaximal exercise: original research. *J Exerc Nutr*. 2024;7(1). doi:10.53520/jen2024.103165

Levac D, Colquhoun H, O'Brien KK. Scoping studies: advancing the methodology. *Implement Sci*. 2010;5:69. doi:10.1186/1748-5908-5-69

Tricco AC, Lillie E, Zarin W, O'Brien KK, Colquhoun H, Levac D, et al. PRISMA extension for scoping reviews (PRISMA-ScR): checklist and explanation. *Ann Intern Med*. 2018;169(7):467–73. doi:10.7326/M18-0850

Pubrica. PCC research question framework [Internet]. 2026 [cited 2026 Apr 20]. Available from: <https://pubrica.com/academy/concepts-definitions/pcc-research-question-framework/>

Stoll CRT, Izadi S, Fowler S, Green P, Suls J, Colditz GA. The value of a second reviewer for study selection in systematic reviews. *Res Synth Methods*. 2019;10(4):539–45. doi:10.1002/jrsm.1369

Polanin JR, Pigott TD, Espelage DL, Grotzinger JK. Best practice guidelines for abstract screening large-evidence systematic reviews and meta-analyses. *Res Synth Methods*. 2019;10(3):330–42. doi:10.1002/jrsm.1354

Raftery J, Young A, Stanton L, et al. Clinical trial metadata: defining and extracting metadata on the design, conduct, results and costs of 125 randomised clinical trials funded by the National Institute for Health Research Health Technology Assessment programme. Southampton (UK): NIHR Journals Library; 2015.

Kovacs EM, Westerterp-Plantenga MS, Saris WH. The effects of 2-week ingestion of (–)-hydroxycitrate and (–)-hydroxycitrate combined with medium-chain triglycerides on satiety, fat oxidation, energy expenditure and body weight. *Int J Obes Relat Metab Disord*. 2001;25(7):1087–94. doi:10.1038/sj.ijo.0801605

Nosaka N, Maki H, Suzuki Y, Haruna H, Ohara A, Kasai M, et al. Effects of margarine containing medium-chain triacylglycerols on body fat reduction in humans. *J Atheroscler Thromb*. 2003;10(5):290–8. doi:10.5551/jat.10.290

Nosaka N, Tsujino S, Honda K, Suemitsu H, Kato K. Enhancement of fat oxidation during submaximal exercise in sedentary persons: variations by medium-chain fatty acid composition and age group. *Lipids*. 2020;55(2):173–83. doi:10.1002/lipd.12222

Nosaka N, Tsujino S, Honda K, Suemitsu H, Kato K, Kondo K. Effect of ingestion of medium-chain triglycerides on substrate oxidation during aerobic exercise could depend on sex difference in middle-aged sedentary persons. *Nutrients*. 2021;13(1):36. doi:10.3390/nu13010036

Roynette CE, Rudkowska I, Nakhasi DK, Jones PJ. Structured medium and long chain triglycerides show short-term increases in fat oxidation, but no changes in adiposity in men. *Nutr Metab Cardiovasc Dis*. 2008;18(4):298–305. doi:10.1016/j.numecd.2006.11.004

- St-Onge MP, Bourque C, Jones PJ, Ross R, Parsons WE. Medium- versus long-chain triglycerides for 27 days increases fat oxidation and energy expenditure without resulting in changes in body composition in overweight women. *Int J Obes.* 2003;27(1):95–102. doi:10.1038/sj.ijo.0802169
- St-Onge MP, Jones PJ. Greater rise in fat oxidation with medium-chain triglyceride consumption relative to long-chain triglyceride is associated with lower initial body weight and greater loss of subcutaneous adipose tissue. *Int J Obes.* 2003;27(12):1565–71. doi:10.1038/sj.ijo.0802467
- Tsuji H, Kasai M, Takeuchi H, Nakamura M, Okazaki M, Kondo K. Dietary medium-chain triacylglycerols suppress accumulation of body fat in a double-blind, controlled trial in healthy men and women. *J Nutr.* 2001;131(11):2853–9. doi:10.1093/jn/131.11.2853
- Valente FX, Cândido FG, Lopes LL, Dias DM, Carvalho SD, Pereira PF, et al. Effects of coconut oil consumption on energy metabolism, cardiometabolic risk markers, and appetitive responses in women with excess body fat. *Eur J Nutr.* 2018;57(4):1627–37. doi:10.1007/s00394-017-1448-5
- Dulloo AG, Fathi M, Mensi N, Girardier L. Twenty-four-hour energy expenditure and urinary catecholamines of humans consuming low-to-moderate amounts of medium-chain triglycerides: a dose-response study in a human respiratory chamber. *Eur J Clin Nutr.* 1996;50(3):152–8.
- Goedecke JH, Clark VR, Noakes TD, Lambert EV. The effects of medium-chain triacylglycerol and carbohydrate ingestion on ultra-endurance exercise performance. *Int J Sport Nutr Exerc Metab.* 2005;15(1):15–27. doi:10.1123/ijsnem.15.1.15
- Misell LM, Lagomarcino ND, Schuster V, Kern M. Chronic medium-chain triacylglycerol consumption and endurance performance in trained runners. *J Sports Med Phys Fitness.* 2001;41(2):210–5.
- Scalfi L, Coltorti A, Contaldo F. Postprandial thermogenesis in lean and obese subjects after meals supplemented with medium- and long-chain triglycerides. *Am J Clin Nutr.* 1991;53(5):1130–3. doi:10.1093/ajcn/53.5.1130
- Tsuchiya Y, Ueda H, Shimizu T, Yokoi K, Yanagimoto K, Ochi E. Eicosapentaenoic acid and medium-chain triacylglycerol structured lipid supplementation improves muscular endurance exercise performance and reduces muscle fatigue in young healthy men. *J Am Nutr Assoc.* 2025;44(3):190–7. doi:10.1080/27697061.2024.2413363
- Coleman H, Quinn P, Clegg ME. Medium-chain triglycerides and conjugated linoleic acids in beverage form increase satiety and reduce food intake in humans. *Nutr Res.* 2016;36(6):526–33. doi:10.1016/j.nutres.2016.01.004
- Courchesne-Loyer A, et al. Stimulation of mild, sustained ketonemia by medium-chain triacylglycerols in healthy humans: estimated potential contribution to brain energy metabolism. *Nutrition.* 2013;29(4):635–40. doi:10.1016/j.nut.2012.09.009
- Dias VC, Fung E, Snyder FF, Carter RJ, Parsons HG. Effects of medium-chain triglyceride feeding on energy balance in adult humans. *Metabolism.* 1990;39(9):887–91.
- Hill JO, Peters JC, Swift LL, Yang D, Sharp T, Abumrad N, et al. Changes in blood lipids during six days of overfeeding with medium- or long-chain triglycerides. *J Lipid Res.* 1990;31(3):407–16.

- Kern M, Lagomarcino ND, Misell LM, Schuster V. The effect of medium-chain triacylglycerols on the blood lipid profile of male endurance runners. *J Nutr Biochem*. 2000;11(5):288–92. doi:10.1016/s0955-2863(00)00081-4
- Kasai M, Nosaka N, Maki H, Negishi S, Aoyama T, Nakamura M, et al. Effect of dietary medium- and long-chain triacylglycerols on accumulation of body fat in healthy humans. *Asia Pac J Clin Nutr*. 2003;12(2):151–60.
- Ogawa A, Nosaka N, Kasai M, Aoyama T, Okazaki M, Igarashi O, et al. Dietary medium- and long-chain triacylglycerols accelerate diet-induced thermogenesis in humans. *J Oleo Sci*. 2007;56(6):283–7. doi:10.5650/jos.56.283
- Stouthard JM, Endert E, Romijn JA, Sauerwein HP. Infusion of long-chain or medium-chain triglycerides inhibits peripheral glucose metabolism in men. *JPEN J Parenter Enteral Nutr*. 1994;18(5):436–41. doi:10.1177/0148607194018005436
- Bickham DC, Bentley DJ, Le Rossignol PF, Cameron-Smith D. The effects of short-term sprint training on MCT expression in moderately endurance-trained runners. *Eur J Appl Physiol*. 2006;96(6):636–43. doi:10.1007/s00421-005-0100-x
- Papamandjaris AA, White MD, Raeini-Sarjaz M, Jones PJ. Endogenous fat oxidation during medium-chain versus long-chain triglyceride feeding in healthy women. *Int J Obes*. 2000;24(9):1158–66. doi:10.1038/sj.ijo.0801350
- Capili B, Anastasi JK. An introduction to the crossover trial design. *Am J Nurs*. 2024;124(9):40–3. doi:10.1097/01.NAJ.0001050812.23977.85
- McMurray RG, Soares J, Caspersen CJ, McCurdy T. Examining variations of resting metabolic rate of adults: a public health perspective. *Med Sci Sports Exerc*. 2014;46(7):1352–8. doi:10.1249/MSS.0000000000000232
- Ahmadi MN, Brookes D, Chowdhury A, Pavey T, Trost SG. Free-living evaluation of laboratory-based activity classifiers in preschoolers. *Med Sci Sports Exerc*. 2020;52(5):1227–34. doi:10.1249/MSS.0000000000000221
- Alexandrou E, Herzberg GR, White MD. High-level medium-chain triglyceride feeding and energy expenditure in normal-weight women. *Can J Physiol Pharmacol*. 2007.
- Angus DJ, Hargreaves M, Dancy J, Febbraio MA. Effect of carbohydrate or carbohydrate plus medium-chain triglyceride ingestion on cycling time trial performance. *J Appl Physiol*. 2000.
- Hill JO, Peters JC, Yang D, et al. Thermogenesis in humans during overfeeding with medium-chain triglycerides. *Metabolism*. 1989.
- Jeukendrup AE, Saris WHM, Brouns F, et al. Effects of carbohydrate and fat supplementation on carbohydrate metabolism during prolonged exercise. *Metabolism*. 1996.
- Jeukendrup AE, Saris WHM, Schrauwen P, et al. Metabolic availability of medium-chain triglycerides coingested with carbohydrates during prolonged exercise. *J Appl Physiol*. 1995.

- Jeukendrup AE, Thielen JJHC, Wagenmakers AJM, et al. Effect of medium-chain triacylglycerol and carbohydrate ingestion during exercise on substrate utilization and subsequent cycling performance. *Am J Clin Nutr*. 1998.
- Kanta JM, et al. Induction of erythropoietin by dietary medium-chain triacylglycerol in humans. *Am J Physiol Endocrinol Metab*. 2025.
- Kasai M, et al. Comparison of diet-induced thermogenesis of foods containing medium- versus long-chain triacylglycerols. *J Nutr Sci Vitaminol*. 2002.
- Nosaka N, et al. Effect of ingestion of medium-chain triacylglycerols on moderate- and high-intensity exercise in recreational athletes. *J Nutr Sci Vitaminol*. 2009.
- Nosaka N, et al. Medium-chain triglycerides with maltodextrin increase fat oxidation during moderate-intensity exercise and extend high-intensity exercise duration. *J Oleo Sci*. 2018.
- Oöpik V, et al. Effects of daily medium-chain triglyceride ingestion on energy metabolism and endurance performance capacity in well-trained runners. *Nutr Res*. 2001.
- Papamandjaris AA, White MD, Jones PJH. Components of total energy expenditure in healthy young women are not affected after 14 days of feeding with medium- vs long-chain triglycerides. *Obes Res*. 1999.
- Thorburn MS, et al. Attenuated gastric distress but no benefit to performance with adaptation to octanoate-rich esterified oils in well-trained male cyclists. *J Appl Physiol*. 2006.
- Van Wymelbeke V, Louis-Sylvestre J, Fantino M. Substrate oxidation and control of food intake in men after meals supplemented with carbohydrate, LCT, or MCT. *Am J Clin Nutr*. 2001.
- Van Zyl CG, Lambert EV, Hawley JA, Noakes TD, Dennis SC. Effects of medium-chain triglyceride ingestion on fuel metabolism and cycling performance. *J Appl Physiol*. 1996.
- Vistisen B, Nybo L, Xu X, Høy CE, Kiens B. Minor amounts of plasma medium-chain fatty acids and no improved time trial performance after consuming lipids. *Am J Clin Nutr*. 2003.